THE HEADLIGHT AUTO LEVELING SYSTEM DOES NOT OPERATE IN RESPONSE TO VEHICLE POSTURE [HEADLIGHT AUTO LEVELING SYSTEM]

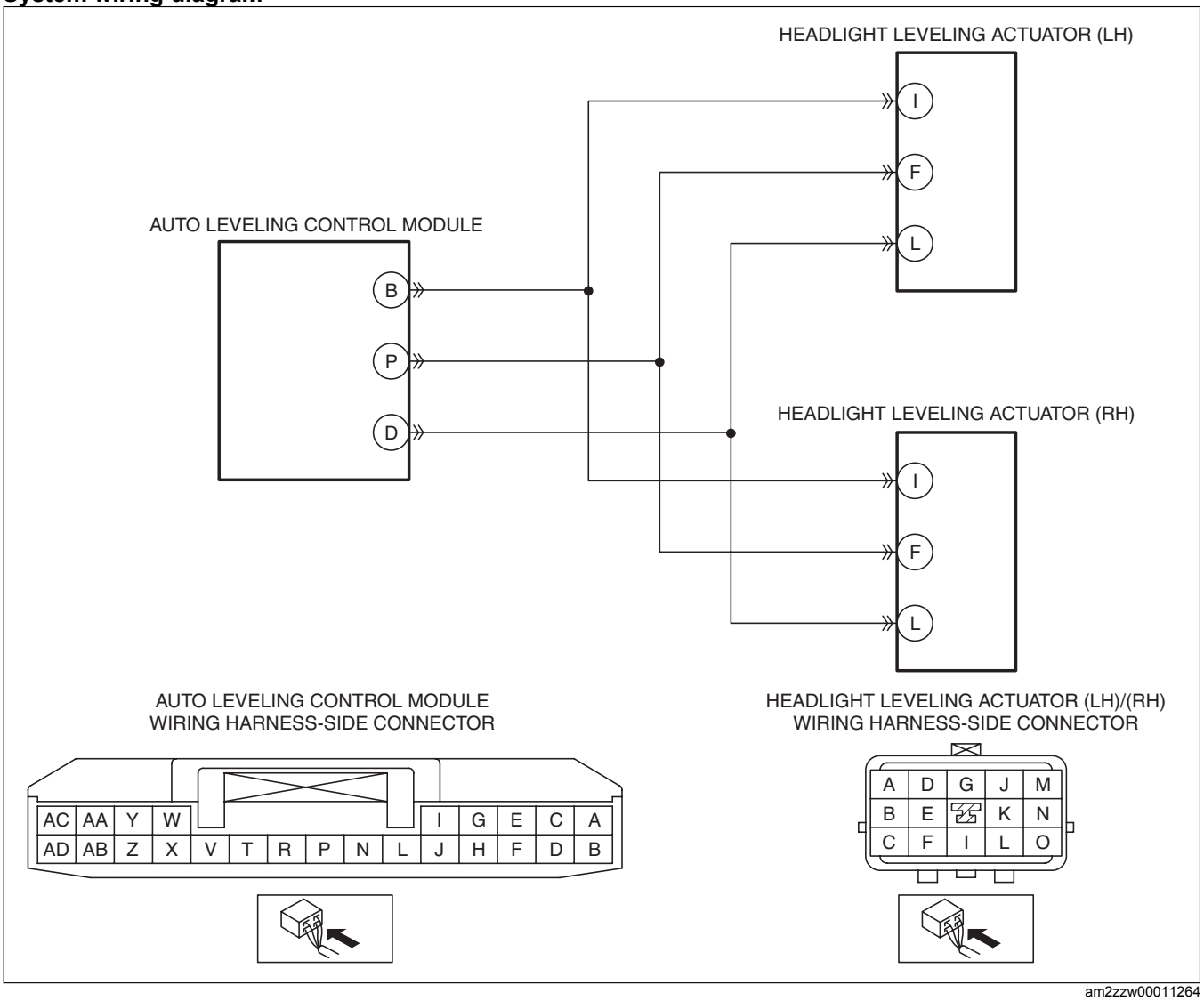
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Note
- If there is any vehicle malfunction complaint lodged by a customer, perform FOREWORD [HEADLIGHT AUTO LEVELING SYSTEM] malfunction diagnosis according to the troubleshooting procedure. (See FOREWORD [HEADLIGHT AUTO LEVELING SYSTEM].)

Description
- The headlight auto leveling system does operate in response to the number of occupants, cargo load in trunk, and the fuel level.
- One side of the headlight leveling actuator does not operate.

Possible malfunction
- Open circuit in wiring harness between auto leveling control module terminal P and headlight leveling actuator (LH)/(RH) terminal F
- Short to ground in wiring harness between auto leveling control module terminal P and headlight leveling actuator (LH)/(RH) terminal F
- Headlight leveling actuator malfunction
- Auto leveling control module malfunction
- Open circuit in wiring harness between the following terminals:
  - Auto leveling control module terminal B—Headlight leveling actuator (LH)/(RH) terminal I
  - Auto leveling control module terminal D—Headlight leveling actuator (LH)/(RH) terminal L
- Short to ground in wiring harness between the following terminals:
  - Auto leveling control module terminal B—Headlight leveling actuator (LH)/(RH) terminal I
  - Auto leveling control module terminal D—Headlight leveling actuator (LH)/(RH) terminal L
- Poor installation of the auto leveling sensor
- Auto leveling sensor malfunction

System wiring diagram



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Diagnostic Procedure

Step	Inspection		Action
1	**DETERMINE IF MALFUNCTIONING LOCATION IS AUTO LEVELING SENSOR** • Verify the malfunction symptom. • Does only one side of the headlight auto leveling system operate?	Yes	Go to the next step.
		No	Go to Step 6.
2	**LEVELING OPERATION SIGNAL INSPECTION** • Using the M-MDS, perform the headlight leveling actuator operation check mode. (See HEADLIGHT LEVELING ACTUATOR OPERATION CHECK MODE [HEADLIGHT AUTO LEVELING SYSTEM]) • Does the headlight leveling actuator operate normally?	Yes	Go to Step 8.
		No	Go to the next step.

Step	Inspection		Action
3	**INSPECT LEVELING OPERATION SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Inspect the wiring harness for continuity between the auto leveling control module and headlight leveling actuator on the side in which the auto leveling system does not operate. — Auto leveling control module terminal P—Headlight leveling actuator (LH)/(RH) terminal F • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify if there is a common connector between auto leveling control module terminal P and headlight leveling actuator (LH)/(RH) terminal F. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location and go to Step 10. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit and go to Step 10.
4	**INSPECT LEVELING OPERATION SIGNAL CIRCUIT FOR SHORT TO GROUND** • Inspect for continuity between the ground and the wiring harness between the auto leveling control module and headlight leveling actuator on the side in which the auto leveling system does not operate. — Auto leveling control module terminal P—Body ground • Is there continuity?	Yes	Refer to the wiring diagram and verify if there is a common connector between auto leveling control module terminal P and headlight leveling actuator (LH)/(RH) terminal F. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location and go to Step 10. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground and go to Step 10.
		No	Go to the next step.
5	**DETERMINE IF MALFUNCTION CAUSE IS HEADLIGHT LEVELING ACTUATOR OR AUTO LEVELING CONTROL MODULE** • Replace the front combination light in which the auto leveling system does not operate. (See FRONT COMBINATION LIGHT REMOVAL/INSTALLATION) • Does the auto leveling system operate correctly?	Yes	Troubleshooting completed.
		No	Replace the auto leveling control module and go to Step 10. (See AUTO LEVELING CONTROL MODULE REMOVAL/INSTALLATION)
6	**INSPECT WIRING HARNESS BETWEEN HEADLIGHT LEVELING ACTUATOR AND AUTO LEVELING CONTROL MODULE FOR OPEN CIRCUIT** • Inspect for continuity in the wiring harness between the following terminals: — Auto leveling control module terminal B—Headlight leveling actuator (LH)/(RH) terminal I — Auto leveling control module terminal P—Headlight leveling actuator (LH)/(RH) terminal F — Auto leveling control module terminal D—Headlight leveling actuator (LH)/(RH) terminal L • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify if there is a common connector between the following terminals. • Auto leveling control module terminal B—Headlight leveling actuator (LH)/(RH) terminal I • Auto leveling control module terminal P—Headlight leveling actuator (LH)/(RH) terminal F • Auto leveling control module terminal D—Headlight leveling actuator (LH)/(RH) terminal L **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location and go to Step 10. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit and go to Step 10.

Step	Inspection		Action
7	**INSPECT WIRING HARNESS BETWEEN HEADLIGHT LEVELING ACTUATOR AND AUTO LEVELING CONTROL MODULE FOR SHORT TO GROUND** • Inspect for continuity between body ground and the following terminals. — Auto leveling control module terminal B — Auto leveling control module terminal P — Auto leveling control module terminal D • Is there continuity?	Yes	Refer to the wiring diagram and verify if there is a common connector between the following terminals. • Auto leveling control module terminal B—Headlight leveling actuator (LH)/(RH) terminal I • Auto leveling control module terminal P—Headlight leveling actuator (LH)/(RH) terminal F • Auto leveling control module terminal D—Headlight leveling actuator (LH)/(RH) terminal L **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for short to ground to determine the malfunctioning location. • Repair or replace the malfunctioning location and go to Step 10. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground and go to Step 10.
		No	Go to the next step.
8	**VERIFY AUTO LEVELING SENSOR INSTALLATION** • Verify the auto leveling sensor installation condition. • Is a bracket bent, is there damage, or is a link disconnected?	Yes	Repair or replace the malfunctioning location and go to Step 10.
		No	Go to the next step.
9	**AUTO LEVELING SENSOR INSPECTION** • Inspect the auto leveling sensor. (See AUTO LEVELING SENSOR INSPECTION) • Is the auto leveling sensor normal?	Yes	Replace the auto leveling control module. (See AUTO LEVELING CONTROL MODULE REMOVAL/INSTALLATION) Go to the next step.
		No	Replace the auto leveling sensor. (See AUTO LEVELING SENSOR REMOVAL/INSTALLATION) Go to the next step.
10	**VERIFY IF MALFUNCTION CAUSE IS CORRECTED** • Switch the ignition to ACC or ON (engine off or on). • Does the LED headlight warning light illumination turn off?	Yes	Troubleshooting completed. (explain the contents of the servicing to the customer)
		No	Verify the malfunction symptom in the symptom troubleshooting chart and perform the other applicable malfunction diagnosis. (See SYMPTOM TROUBLESHOOTING [HEADLIGHT AUTO LEVELING SYSTEM].)